

ZETA FUNCTIONS OF SIMPLE ALGEBRAS

AND ADAPTATION OF THE BOOK BY
JACQUET AND GODEMENT

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Chapter

INTRODUCTION

Chapter I

THE LOCAL THEORY

SECTION 1

CONVERGENCE LEMMAS

In this section, we establish the convergence of some integrals which will be needed later. Let F be a (commutative) local field and H a division algebra with center F . Let M be the simple algebra of $m \times m$ matrices with entries in H . If d^2 is the rank of H over F , then rank of M over F is n^2 , where $n = md$.

EXAMPLE 1.1 (Prototypical Example). The simplest non-trivial example of the above setup would be when $H = F$, and

$$M = \text{Mat}_{n \times n}(F), \quad M^\times = \text{GL}_n(F).$$

We recall some terminology and properties about central simple algebras.

DEFINITION 1.2. Let F be a field, and A a central simple F -algebra (i.e A is a finite-dimensional associative F -algebra whose center is F). A finite extension K/F is said to be a **splitting field for A** if

$$K \otimes_F A \cong \text{Mat}_{n \times n}(K)$$

for some n . We take it as a fact here (which is not too hard to show) that every such A admits a splitting field.

As a consequence, if we let K be the splitting field of H over F and G the unit group of M , the above definition/fact tells us that any such G is of the form $\text{Mat}_{m \times m}^\times(H)$, and thus base changing to K gives:

$$G_K \cong \text{Mat}_{m \times m}^\times(K \otimes_F H) \cong \text{GL}_m(\text{Mat}_{k \times k}(K)).$$

Thus we deduce that any such G is a **reductive** algebraic group.

DEFINITION 1.3. With the setup as above, the **reduced norm**, denoted $\nu_M : M \rightarrow F$, is the unique polynomial maps such that the following diagram commutes:

$$\begin{array}{ccc}
M & \xrightarrow{\nu_M} & F \\
\downarrow & & \downarrow \\
K \otimes M & \xrightarrow{\nu_M \otimes \mathbb{1}} & K \\
\downarrow \wr & & \parallel \\
\mathrm{Mat}_{k \times k}(K) & \xrightarrow{\det} & K.
\end{array}$$

The **reduced trace** $\tau_M : M \rightarrow F$ is defined similarly.

PROPOSITION 1.4. Let ϕ be in $\mathcal{S}(M)$ and $d^\times x$ a Haar measure on $G(F)$. Then the integral

$$\int_{G(F)} \phi(x) |\nu_M(x)|^s d^\times x$$

is absolutely convergent for $\Re(s) > d(m-1)$.

Proof. Since H is a central simple algebra over F , we may identify the center $Z(G(F))$ with $F^\times$. By virtue of ϕ being a Schwartz-Bruhat function, we may suppose $\phi \geq 0$ and $s \in \mathbb{R}$. Then the integral in the proposition statement is equal to

$$\int_{G(F)/Z(G(F))} |\nu_M(\bar{g})|^s d\bar{g} \int_{F^\times} \phi(ga) |a|^{ns} d^\times a.$$

First assume $m = 1$...

Assume now $m = 1$

□

SECTION 2

INDUCED REPRESENTATIONS

Throughout this section, F is a non-Archimedean local field, and M is a semi-simple algebra over F such that

$$M \cong \bigoplus_i M_i,$$

where each of the M_i are central simple F -algebras.

The multiplicative group G of such an M as above is a reductive algebraic group defined over F (whose R -rational points we denote $G(R)$ for any F -algebra R , though we often just write G if context is clear) and clearly we have

$$G \cong \prod_i G_i, \tag{2-1}$$

where $G_i := M_i^\times$.

We are interested in representations π of $G(F)$ – henceforth denoted as just G – on a $\mathbb{C}$ -vector space V that are *smooth and admissible*.

DEFINITION 2.1. A representation (π, V) of G is said to be **smooth** if $\text{Stab}_G(v)$ is an open subgroup of G for every $v \in V$. It is **admissible** if it is smooth and V^H (the space of vectors in V stabilized by H) is finite dimensional for every compact open subgroup H .

REMARK. One can verify that smoothness of a representation is equivalent to continuity of the map

$$\begin{aligned} G \times V &\rightarrow V \\ (g, v) &\mapsto \pi(g)v \end{aligned}$$

if we give V the trivial locally convex topology in which every seminorm is continuous.

DEFINITION 2.2. We denote by $\mathcal{H}(G(F))$ the space of locally constant and compactly supported complex-valued functions on $G(F)$. Convolution turns this into a complex algebra, known as the **Hecke algebra**.

Suppose (π, V) is a smooth representation of G . Then for any $f \in \mathcal{H}(G)$ we may define an operator $\pi(f)$ on V in the following manner:

$$\pi(f) : v \mapsto \int_G f(g) \pi(g)v \, dg. \quad (2-2)$$

Note by assumption f is locally constant and compactly supported, so the above integral is in fact a finite linear combination of vectors in V . Thus we may also view $\pi : f \mapsto \pi(f)$ as a representation of $\mathcal{H}(G)$ on V .

If we denote by λ the shift operator on $\mathcal{H}(G)$, i.e

$$\lambda(g)f(h) := f(g^{-1}h),$$

then we note we have the identity

$$\begin{aligned} \pi(\lambda(g)f)v &= \int_G f(g^{-1}h) \pi(h)v \, dh \\ &= \int_G f(h) \pi(hg)v \, dh \\ &= \int_G f(h) \pi(g) \pi(h)v \, dh \\ &= \pi(g) \pi(f)v. \end{aligned}$$

LEMMA 2.3. A representation (π, V) of G is smooth if and only if the induced representation π on $\mathcal{H}(G)$ satisfies that for every $v \in V$ there exists some $f \in \mathcal{H}(G)$ so that $\pi(f)v = v$.

Proof. If π is smooth, then by assumption $\text{Stab}_G(v)$ is an open (hence closed!) subgroup. We may pick f then to be an appropriate multiple of the characteristic function of a compact open set C within this open subgroup (specifically it should be multiplied by the reciprocal of the Haar measure of the compact open), which would immediately give

$$\pi(f)v := \int_G f(g) \pi(g)v \, dg = \frac{1}{\text{Vol}(C)} \int_C \pi(g)v \, dg = \frac{1}{\text{Vol}(C)} \cdot \text{Vol}(C)v = v.$$

Conversely, suppose we have a representation (π, V) of $\mathcal{H}(G)$ so that every vector v has some f so that $\pi(f)v =$

v . We will construct a smooth (π', V) representation of G so that

$$\pi(f)v = \int_G f(g)\pi'(g)v \, dg$$

for every $v \in V$.

□

DEFINITION 2.4. An irreducible admissible representation

SECTION 3

REDUCTION TO THE ABSOLUTELY CUSPIDAL CASE

We keep the notation of the previous section but we assume that M here is a simple algebra. In particular, we can take

$$G = \mathrm{GL}_m(H),$$

where H is a central division algebra over F . We note that the center of G may be identified with $F^\times$. As always by abuse of notation we denote by M or G

We denote by $\mathcal{S}(M)$ the space of compactly supported, locally constant functions from M to $\mathbb{C}$. We denote by τ, ν the reduced trace and reduced norm respectively.

DEFINITION 3.1. Let ψ be a nontrivial additive character of a local field F , and M a simple matrix algebra over F . The **Fourier transform** of a function $\phi \in \mathcal{S}(M)$ is

$$\widehat{\phi}(x) := \int_M \phi(y) \psi(\tau(xy)) \, dy.$$

Here the Haar measure dy is chosen so that $\int_M \widehat{\phi}(x) \, dx = \phi(0)$.

Need to know that the trace bilinear form is non-degenerate, i.e the map.

DEFINITION 3.2. Let (π, V) be an admissible representation of G . A **coefficient** of π is a function f of the form

$$f(g) = \langle \pi(g)v, \tilde{v} \rangle,$$

where $v \in V$ and ...

SECTION 4

EXAMPLE: SPHERICAL FUNCTIONS

Again we assume that

$$G = \mathrm{GL}_m(H),$$

where H is a division algebra of center $F^\times$ and rank d^2 . Let

$$K := \mathrm{GL}_m(\mathcal{O}_H),$$

which is a maximal compact subgroup of G . We may assign K to have measure one.

Assume first $m = 1$. Then K is the group G_0 of (reduced) norm 1 elements of G . If π is an irreducible representation of G that contains the unit representation of G_0 , it must have the form

$$\pi = \chi \circ \nu,$$

where χ is an unramified quasicharacter of $F^\times$. We have seen in this case that

$$L(s, \pi) = L(s, \chi\mu_1)$$

Chapter II

GLOBAL THEORY

SECTION 5

AUTOMORPHIC FORMS

In this chapter K will be a global field, H will be a division algebra of center F and rank d^2 , M will be the simple algebra $\text{Mat}_{m \times m}(H)$, and G will be group $M^\times$. We will denote by $\mathbb{A}_K$ and $\mathbb{I}_K$ the adeles and ideles of K respectively. For each place $v \in \mathcal{P}_K$ we denote by K_v the (local!) completion of K at v and put

$$H_v := H \otimes_K K_v, \quad M_v := M \otimes_K K_v, \quad G_v := \text{GL}_m(H_v).$$

For each v we choose an isomorphism

$$H_v \xrightarrow{\sim} \text{Mat}_{m'_v \times m'_v}(H'_v), \tag{5-1}$$

where H'_v is a division algebra of center K_v and rank $(d'_v)^2$, and $d = m'_v d'_v$. From 5-1, we derive isomorphisms

$$M_v \xrightarrow{\sim} \text{Mat}_{mm'_v \times mm'_v}(H'_v), \quad G_v \xrightarrow{\sim} \text{GL}_{mm'_v}(H'_v). \tag{5-2}$$

The isomorphism in (5-1) – and therefore the isomorphisms in 5-2 – are defined up to an inner automorphism. In particular, our choice of isomorphism does not affect the representation theory.

For each $v \in \mathcal{P}_K$ we also define the constant

$$a_v := (-1)^{mm'_v(d'_v-1)} \tag{5-3}$$

Note that for almost all (i.e. cofinitely many) v we have

$$H_v = \text{Mat}_{d_v \times d_v}(K_v), \quad H'_v = K_v, \quad M_v = \text{Mat}_{n \times n}(K_v), \quad G_v = \text{GL}_n(K_v),$$

with

$$d'_v = 1, \quad m'_v = 1, \quad a_v = 1.$$

It is an easy consequence of the reciprocity law (in the Hasse form) that

$$\prod_{v \in \mathcal{P}_K} a_v = 1.$$

We shall give a direct proof of this later [Where?].

Using the identification of G_v in (5-2), we define for each v a compact subgroup $H_v \leq G_v$. If v is Archimedean,

H_v is the group of the §8, and if v is non-Archimedean then H_v is the group

$$\mathrm{GL}_{mm'_v}(R'_v),$$

where R'_v is the ring of integers in H'_v .

The product

$$H := \prod_{v \in \mathcal{P}_K} H_v$$

is a maximal compact subgroup of the adelic group

$$G(\mathbb{A}_K) = \mathrm{GL}_m(H(\mathbb{A}_K)).$$

We denote by G_0 the subgroup of $g \in G(\mathbb{A}_K)$ such that $|\nu_M(g)| = 1$.

Exactly as in [Insert source here](#) we define the

DEFINITION 5.1. A continuous function ϕ on $G(K) \backslash G(\mathbb{A}_K)$ is said to be **cuspidal** if the integral

$$\int_{U_K \backslash U(\mathbb{A}_K)} \phi(ug) \, du$$

vanishes for all U that are unipotent radicals of a proper K -parabolic subgroup of G .

Let T be a maximal K -split torus of G and α_i (for $i \in [1, m-1]$) a system of simple roots of T . We put

$$\eta(t) := \inf_i |\alpha_i(t)|, \quad \kappa(t) := \sup_i |\alpha_i(t)|.$$

DEFINITION 5.2. A function ϕ on $G(K) \backslash G(\mathbb{A}_K)$ is said to be **slowly increasing** (resp. **rapidly decreasing**) if for any compact $C \subseteq G$ and any $c > 0$ we have $p \geq 1$ and $c' > 0$ such that the relations

$$g \in C, \, t \in T(\mathbb{A}_K) \cap G_0, \, \eta(t) \geq c$$

imply

$$|\phi(tg)| \leq c' \kappa(t)^p \text{ (resp. } |\phi(tg)| \leq c' \kappa(t)^{-p})$$

Whether these condition is satisfied or not is independent of our choice of maximal torus T or simple roots $\{\alpha_i\}$.

When $m = H$, then being slowly increasing is equivalent to being rapidly decreasing, and both conditions are equivalent ϕ being bounded on every compact subset of $G(\mathbb{A}_K)$. Since $(Z(\mathbb{A}_K)G(K) \backslash G(\mathbb{A}_K))$ is compact, then it is sufficient that ϕ is continuous and there is a Hecke character $\chi : \mathbb{I}_K / K^\times$ such that

$$\phi(ag) = \chi(a)\phi(g) \, \forall a \in \mathbb{I}_K, g \in G(\mathbb{A}_K)$$

Let X be a closed subset of $\mathbb{I}_K$ such that the map $x \mapsto |x|_{\mathbb{I}_K}$ is proper. Let also C be a compact subset of $G(\mathbb{A}_K)$ and C' a compact subset of $U_0(\mathbb{A}_K)$. Finally, let c be a strictly positive constant.